Exploratory Data Analysis Report

I. Introduction

This report documents the exploratory data analysis (EDA) conducted for our project on uncovering trust and transparency issues in the Bay Area auto repair industry. Auto repair shops are among the most frequently cited businesses for consumer complaints, concerns around overcharging, unnecessary repairs, and poor communication are widespread. By analyzing customer reviews, we aim to identify measurable signals of these issues at scale.

Our original project proposal identified the Yelp Open Dataset as the primary data source. However, during the data collection phase we discovered a critical limitation: the Yelp Open Dataset does not include businesses in San Francisco or the broader Bay Area. The dataset covers only a fixed set of cities including Atlanta, Austin, Boston, Boulder, Columbus, Orlando, Portland, and Vancouver, none of which are located in California. Because our research question is specifically focused on the Bay Area auto repair market, using Yelp data from an unrelated city would fundamentally undermine the validity of our findings. We therefore pivoted to the Google Places API, which allowed us to directly query auto repair businesses and their customer reviews within the San Francisco Bay Area.

This change also aligns with our professor's feedback on the proposal: that our data pipeline should be general enough to work with similar datasets from other geographic areas. The Google Places API is not geographically constrained, the same pipeline can be re-run for any city or region simply by changing the location parameter, making our approach more flexible and generalizable than the original Yelp-based design.

II. Dataset Description

The dataset contains 971 reviews across 196 unique auto repair businesses in the San Francisco Bay Area. Each row represents a single customer review.

Column	Type	Description
place_id	Categorical (string)	Unique Google Places identifier for each business
business_name	Categorical (string)	Name of the auto repair shop
review_rating	Integer (1–5)	Numeric star rating given by the reviewer
review_text	String (text)	Full text of the customer review
review_time	Categorical (string)	Relative time label (e.g., '2 months ago')
author	String	Name of the reviewer

Two additional features were engineered from the raw data to enable richer analysis:

- **Review_length:** The number of words in the review text, computed using a whitespace-based split. This allows us to measure verbosity as a function of sentiment and rating.
- **Sentiment_score:** The VADER compound sentiment score for each review text, ranging from -1.0 (most negative) to +1.0 (most positive). VADER (Valence Aware Dictionary and sEntiment Reasoner) is a lexicon-based tool well-suited to short, informal consumer text.

III. Data Preprocessing

Before analysis, the dataset was examined for quality issues. The following steps were performed:

- A null check **df.isnull().sum()** revealed that only the **review_text column** contained missing values, 6 rows out of 971 had no text content. All other columns were complete. Because text is essential for our key engineered features **(review_length and sentiment_score)**, rows with missing review_text were retained but treated as empty strings during feature engineering, resulting in a **review_length** of 0 and a neutral **sentiment_score** of 0.0 for those entries.

Column	Missing Values
place_id	0
business_name	0
review_rating	0
review_text	6
review_time	0
author	0

- A duplicate check **df.duplicated().sum()** confirmed that no duplicate rows exist in the dataset. With 965 unique review texts out of 971 total entries, the small number of non-unique text values is attributable to the 6 missing (NaN) entries, not to true duplicate reviews. No rows were removed.

IV. Feature Engineering

After cleaning, two features were created:

- **review_length:** df['review_length'] = df['review_text'].astype(str).apply(lambda x: len(x.split()))
- **sentiment_score:** Applied VADER's SentimentIntensityAnalyzer to each review text, extracting the compound score.

V. Exploratory Data Analysis

The review_rating column exhibits a strong positive skew, which is characteristic of voluntary online review platforms:

Statistics	Value
Count	971
Mean	4.61 ★
Median (50th percentile)	5.0 ★
Std. Deviation	1.14
Minimum	1 ★
Maximum	5 ★
25th percentile (Q1)	5.0 ★
75th percentile (Q3)	5.0 ★

- The median and both quartile values are all 5.0, meaning at least 75% of reviews in the dataset carry a 5-star rating. This reflects the well-documented 'J-curve' effect in online review platforms, where satisfied customers and dissatisfied customers are more likely to leave reviews than neutral ones. The mean of 4.61 is pulled down by a small but meaningful cluster of 1-star reviews (approximately 8.4% of the dataset), which represent the primary signal for trust and transparency concerns.

The distribution of review length (in words) shows high variance, suggesting that some reviewers leave brief comments while others write detailed accounts:

Statistics	**Value**
Mean	76.6 words
Median	59 words
Std. Deviation	72.7 words
Minimum	1 word
25th percentile (Q1)	37 words
75th percentile (Q3)	90 words
Maximum	818 words

- The mean (76.6 words) is substantially higher than the median (59 words), confirming a right skew, a small number of very long reviews pulls the average up. This is relevant to our research question: if dissatisfied customers write longer, more detailed reviews, that verbosity may encode the specific grievances (e.g., overcharging, poor communication) we are trying to extract.

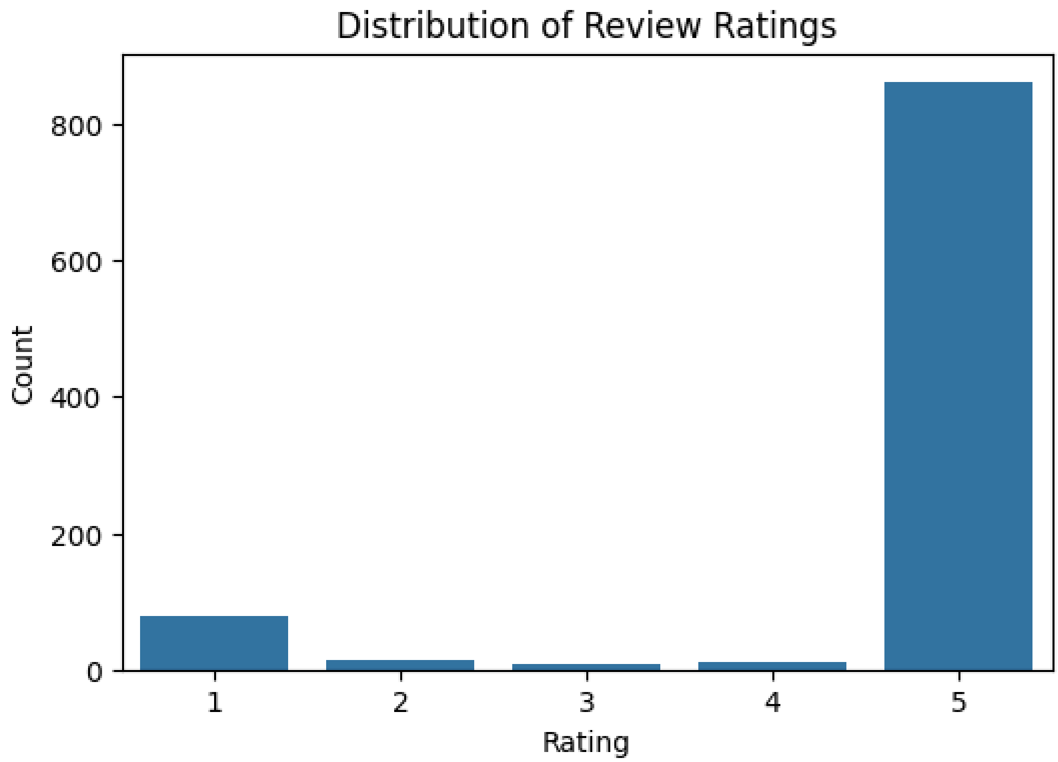

Figure 1 shows the count of reviews at each star rating level. The overwhelming majority of reviews are 5-star, with a sharp drop to 4-star and below. The distribution is heavily left-skewed (most ratings are at the maximum), consistent with the descriptive statistics above.

Interpretation: The dominance of 5-star reviews confirms positive selection bias in Google Places reviews. However, the 1-star and 2-star reviews, though minority, are the most valuable for our downstream analysis, as they are likely to contain explicit complaints about trust and transparency violations.

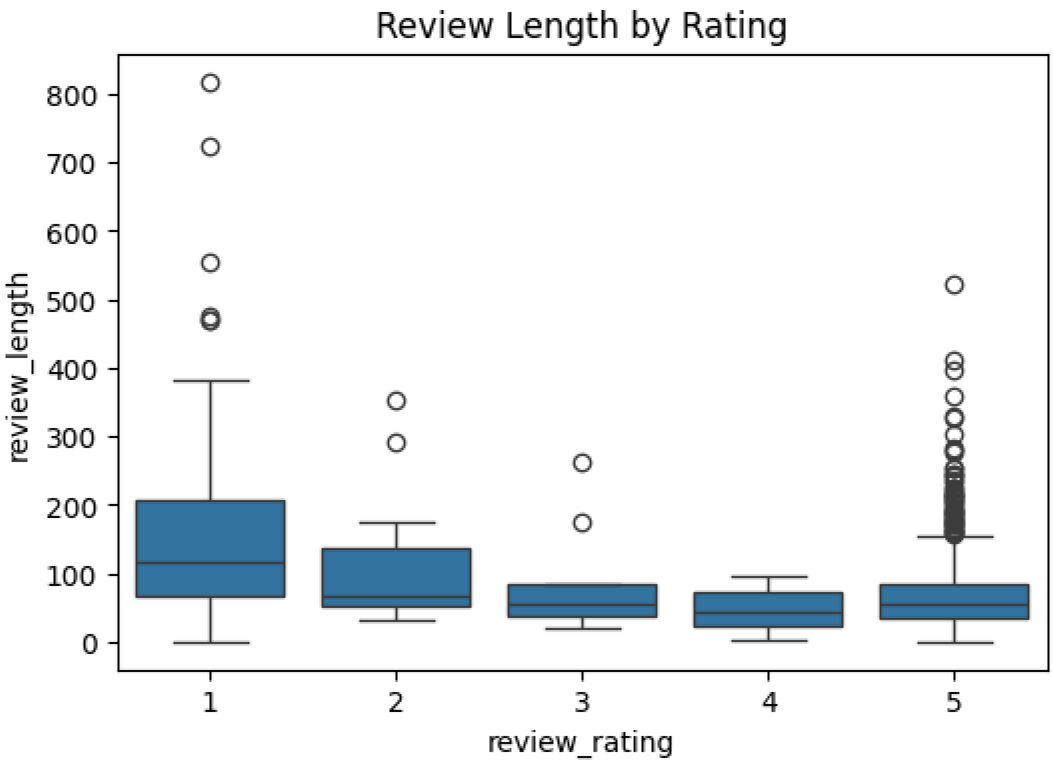

Figure 2 shows box plots of review length (word count) grouped by star rating. A clear trend is visible: lower-rated reviews tend to be longer than higher-rated ones. The median review length for 1-star reviews is noticeably higher than for 5-star reviews. The spread (IQR) is also wider for 1-star reviews. Interpretation: This supports a core hypothesis of our project, customers who feel wronged or deceived tend to write more detailed accounts. The inverse correlation between rating and review length ($r \approx -0.35$ based on the correlation matrix) provides quantitative support for this pattern. Longer negative reviews are likely to contain rich qualitative data about specific complaints, which will be valuable for the next phase of our analysis.

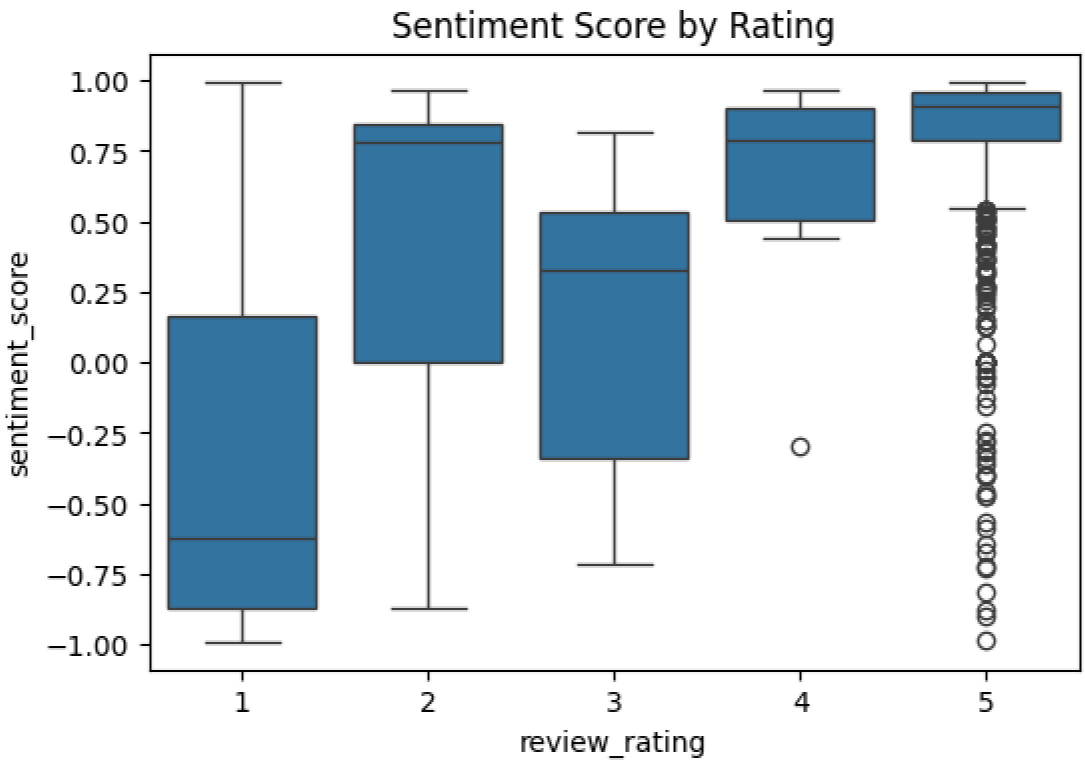

Figure 3 shows box plots of VADER sentiment scores grouped by star rating. A strong monotonic relationship is visible: as star rating increases, sentiment score also increases. 5-star reviews cluster near +1.0 (strongly positive), while 1-star reviews cluster near -0.5 to 0.0 (negative to neutral). Interpretation: The strong positive correlation between rating and sentiment ($r \approx 0.65$) validates VADER as a useful proxy for rating-based sentiment in this domain. Notably, even 1-star reviews do not always score at the extreme negative end of the VADER scale, this may reflect reviews that are factual and measured in tone rather than overtly hostile, which is common in consumer complaints about service issues.

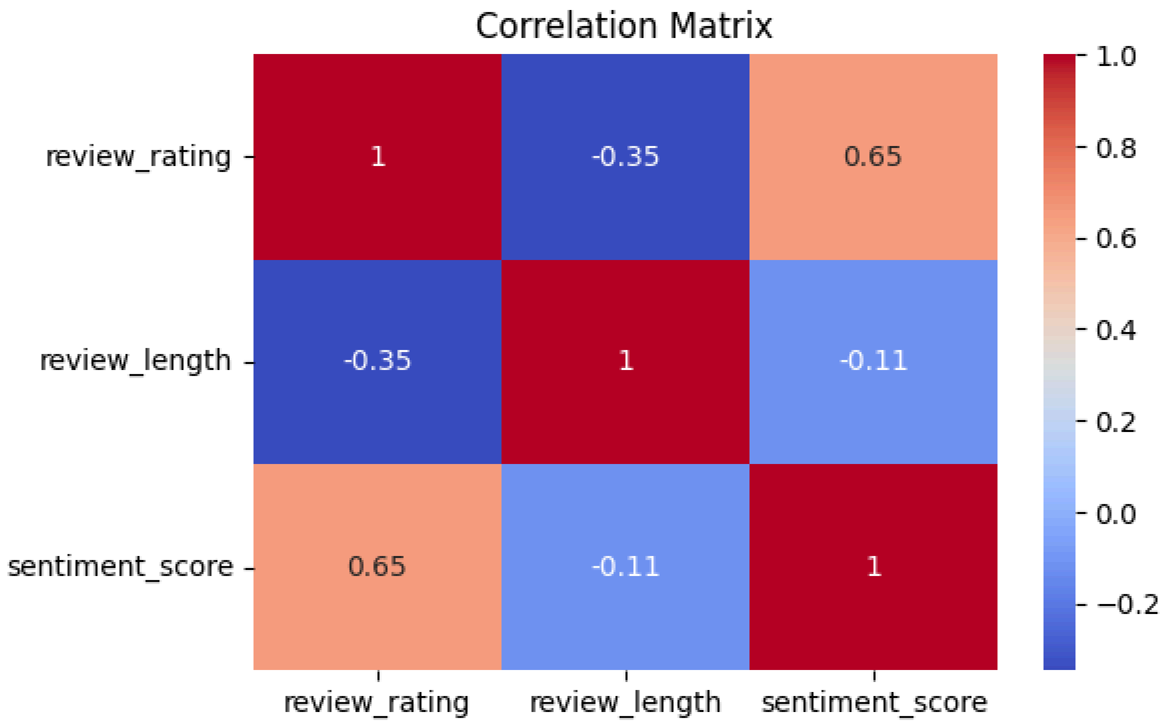

Figure 4 shows the Pearson correlation matrix between review_rating, review_length, and sentiment_score.

Variable Pair	Correlation(r)	Interpretation
review_rating ↔ sentiment_score	≈ +0.65	Strong positive: VADER tracks star ratings well
review_rating ↔ review_length	≈ -0.35	Moderate negative: lower ratings have longer reviews
sentiment_score ↔ review_length	≈ -0.20	Weak negative: more negative reviews tend to be longer

- **Summary of EDA findings**

The EDA has produced four primary insights that will directly shape the next phase of the project:

- Strong Positive Rating Skew: 75%+ of reviews are 5-star, confirming selection bias. Negative reviews are rare but likely information-dense.
- VADER Validates as Sentiment Proxy: The strong correlation ($r \approx 0.65$) between VADER sentiment score and star rating confirms that automated sentiment analysis can reliably track customer satisfaction without manual labeling.
- Dissatisfied Customers Write More: The negative correlation between rating and review length ($r \approx -0.35$) supports the hypothesis that negative reviews are more detailed and more likely to contain specific complaints about pricing, communication, and transparency.
- Trust and Transparency Signals Are Detectable: Even at this exploratory stage, the data structure, rich review text, ratings, and temporal ordering, is appropriate for the downstream analysis we have planned, including keyword extraction and topic modeling on low-rated reviews.